

	WEEKLY PLAN	
Lec	Topics to be covered	Ex
1	Equation of a line, various forms of equation of a line, line segment	12.5
2	Vector and scalar equation of a plane, coordinate planes	12.5
3	Sphere, traces, cylinders	12.6
4	Quadric surfaces	12.6
5	Introduction of plane and space curves and $\mathbf{r}(t)$, helix	13.1
	Derivative of $\mathbf{r}(t)$	13.2
6	Arc length of parametric curves, Arc length function, ds derivation, unit tangent definition	13.3
7	Functions of several variables, domain, graph, level curves, contour map, level surfaces	14.1
8,9	Limit, and continuity of functions of several (two, three and more) variables, composition of function	14.2
10	Partial derivatives, its interpretation, higher derivatives, Laplace, and wave equations.	14.3
11,12	Tangent planes and linear approximations, differentials	14.4
13	Chain rule of differentiation of function of two and three variables, implicit differentiation.	14.5
14	Directional derivatives and the gradient vector	14.6
15,16	Maximum and Minimum Values	14.7
17	Double integrals over rectangles, volume, average value	15.1
18	Double Integrals over General Regions (Type I, Type II etc.), Changing the order of integration, $A(D)$ using double integral	15.2
19	Double Integrals in Polar Coordinates, Jacobian	15.3
20	Surface area of $z = f(x, y)$	15.5
21	Triple Integrals (x -simple, y -simple, z -simple, projections)	15.6

22	Definition of vector field in R^2 and R^3 , Recall gradient of a scalar function, definition of conservative vector field	16.1
23 24	Parametrization of well-known curves, Line integral of $f(x,y)$ over a plane curve C with respect to s , x and y respectively, line integral of $f(x,y,z)$ over a space curve C with respect to s , x , y and z respectively, line integrals of vector fields, Work	16.2
25	The fundamental theorem for line integrals, work done, conservative vector fields and potential functions	16.3
26	Simple curve, closed curve, simple closed curve, connected region, simply connected region, positive orientation, negative orientation, Green's Theorem	16.4
27	Curl and divergence of a vector field, theorem for checking whether a field is conservative ? irrotational vector field, incompressible vector field	16.5
28	Parametric surfaces and their areas, Surface integrals.	16.6, 16.7
29 30	Stokes' Theorem, The Divergence Theorem	16.8, 16.9